

Check Point Task - Food Ordering

TESTING REVIEW & V2

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Testing

I would say overall the testing went very well as there was only 1 exceptional test that failed, though it didn't crash the program but had an undesired outcome. I believe this is because I had a clear idea of what I wanted my program to do and put a lot of focus into making every input guarded, made sure any invalid input was caught and enabled a retry loop with `while (true)` at each of these to allow the user to make the attempt again and not have to exit the program, something I didn't do previously with the calculations exercise and so it was something I wanted to improve on. I made good use of `try/catch` statements and guards, especially for number ranges. As well as this, I used `IF ELSE` statements whenever it was a 1 or 2 choice. I will say it was time consuming to put these in and definitely made my code a lot longer than it had to be to fulfil the requirements for this task, but it was good to see the final product and see it passing almost all of the tests and never crashing, and really once I had done 1 retry loop with guards and `try/catch`, it got a lot easier to figure out the rest. It seems important to me to have my code be robust and consistent rather than flashy and so it was a goal of mine to make sure this was the case before starting the project.

Improvements

My program definitely could be improved and made more efficient though, which I plan to do by making a V2 when the semester is over. The main improvement I would think is the length of the code could be trimmed down. Using methods seemed to make my code a lot more efficient so I will try and make helper methods for a lot more program, in particular with the starters, mains, desserts and drinks sections. The whole point of methods is to make code reusable, and these sections are all very similar, so if I can manage to reuse these sections, I will trim down the length significantly, but I'll need to figure out how to make the minor adjustments for each section like what is printed. I could also make a helper method for retry loops as a lot of them are similar too, but again, I'll need to make minor changes with each one as they're slightly different, but if I can manage this, I think it will make my code a lot more professional. Another thing I want to work on is that currently, it's all in one class, so I think a V2 will be a good opportunity to learn more about classes and see if I can incorporate these to make the code better as well. I did use multiple in the original program, but it was just different versions after I completed the sections to make sure when I added a new part and it didn't work, I still had the original working code. There's lots of other things I want to try too such as using the specific exception types instead of the general `Exception e`, adding a GUI which we are working on currently, making the menu printing section better by making it a single array and printing specific elements. I could make the allergies section more detailed and accurate, maybe by adding them to an `ArrayList` and covering multiple dishes, as well as allowing a user to order an item they've said they're

allergic to if multiple orders are being made, as they might be getting it for someone else, improving the name section to allow names like McDonald etc., and much more. One final improvement I will note, not with the program, but it is that the pseudocode was quite frustrating to make in Word and after looking online, it seems there are better places to do this, such as Notepad so I will make sure that for my next project, I will write the pseudocode using that.